

Mathematical Models Semester Project

Overview

High Level Goals:

The high-level goals for this project are:

- Prepare students for careers in industry.
- Improve students' abilities in the following essential skills:
 - Critical Thinking Skills - includes creative thinking, innovation, inquiry and analysis, evaluation and synthesis of information.
 - Communication Skills - includes effective development, interpretation, and expression of ideas through written, oral, and visual communication.
 - Empirical and Quantitative Skills - includes the manipulation and analysis of numerical data or observable facts resulting in informed conclusions and accurate predictions.
- Groups create their own models based on real or realistic data and scenarios.
- Groups must analyze the created models using sound modeling principles.
- The models must be represented graphically and in terms of mathematical formulas.
- Groups will apply modeling techniques to produce qualitative solutions, make predictions, or explain the data.
- Groups must verify the correctness of their models and discuss possible sources of error.
- Students will write individual reflections on how learning modeling skills assists in the overall process of learning and how these skills may contribute to success in their future careers.

Each student will join or be added to a group from a list provided in Canvas. The group size ranges between three to five. Once groups are set, groups will decide on their choice of Projects: Groups may choose from UMAP modules or ILAP projects (supplements to the text) or suggest a substantial mathematical modeling project of their own interest. No two groups may do the same project topic in the same semester. **Because all projects will concern different topics and possibly different in nature (e.g. empirical model versus proof), groups may not have identical deliverables. Therefore, all groups must meet with the Business Customer (the Instructor) and ensure that their proposed specific deliverables are aligned with the Business Customer's expectations.**

Deliverables

There will be five deliverables for this Project. Three of the deliverables are group work: Group Contract, Part 1 and Part 2. Two deliverables: the Peer Review and the Project Reflection are completed by each individual student.

Group Contract

Submit one group contract for the entire group. The contract outlines the proper behavior of a group member as well as the consequences for breaking these terms. The completed, signed contract will need to be uploaded as a pdf document.

Part 1: The Group's Proposal

Submit one set of documents for the entire group. There are several required parts to Part 1, details are documented separately in the Proposal Requirements document. Submission will be made electronically through Canvas. Follow the instructions on the Semester Project Part 1 page found in the list of assignments of the Syllabus tab in Canvas. The group will also be required to schedule a review meeting with the business customer. (Probably the Instructor.)

Peer Evaluation Form

To receive credit for the Part 1 group submission all students must individually complete and submit a Peer Evaluation Form. **If a student fails to submit the required Peer Evaluation Form, then they will receive a 0 for their Part 1 grade. The Peer Evaluation must be serious and meaningful and not just pro forma.**

Part 2: The Group's Results

Submit one set of deliverables for the entire group. There are several required parts to Part 2, details are documented separately in the Results Requirements document. Submission will be made electronically through Canvas. Follow the instructions on the Semester Project Part 2 page found in the list of assignments of the Syllabus tab in Canvas. The group will also be required to coordinate with the instructor to schedule a presentation of their results to the class.

Project reflection:

Each student will submit an Individual Reflection write-up. Submission will be made electronically through Canvas. Follow the instructions on the Semester Project Reflection page found in the list of assignments of the Syllabus tab in Canvas.

